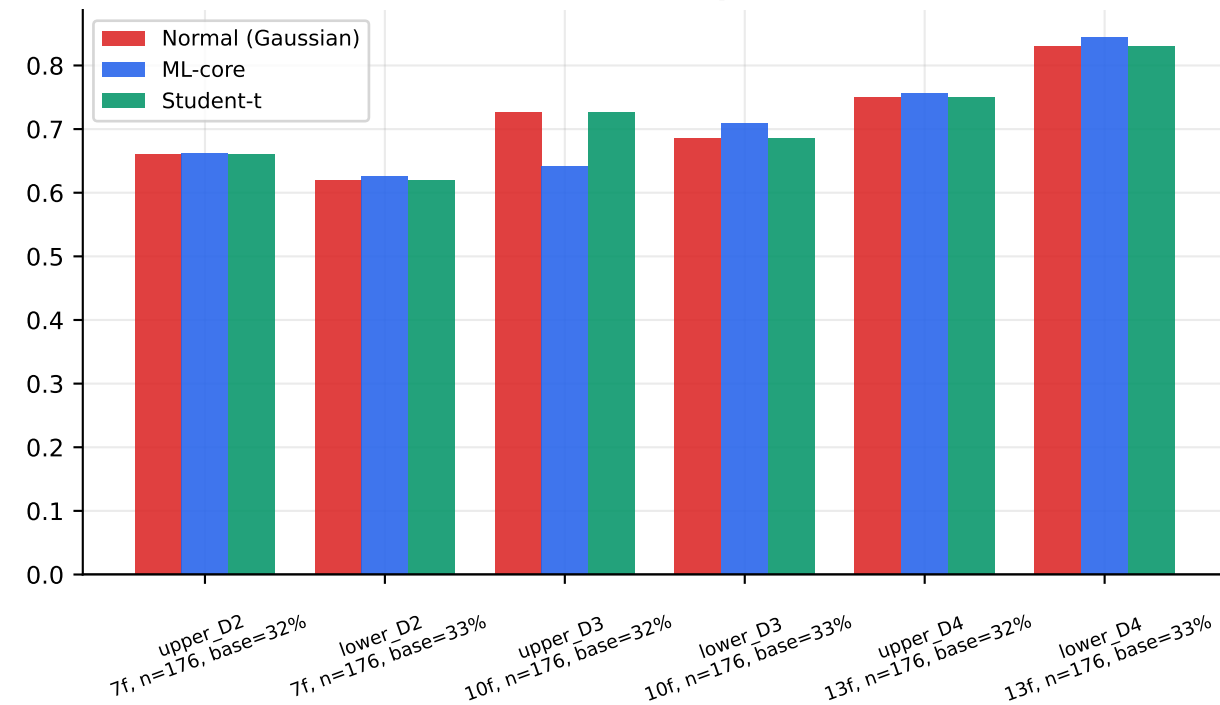
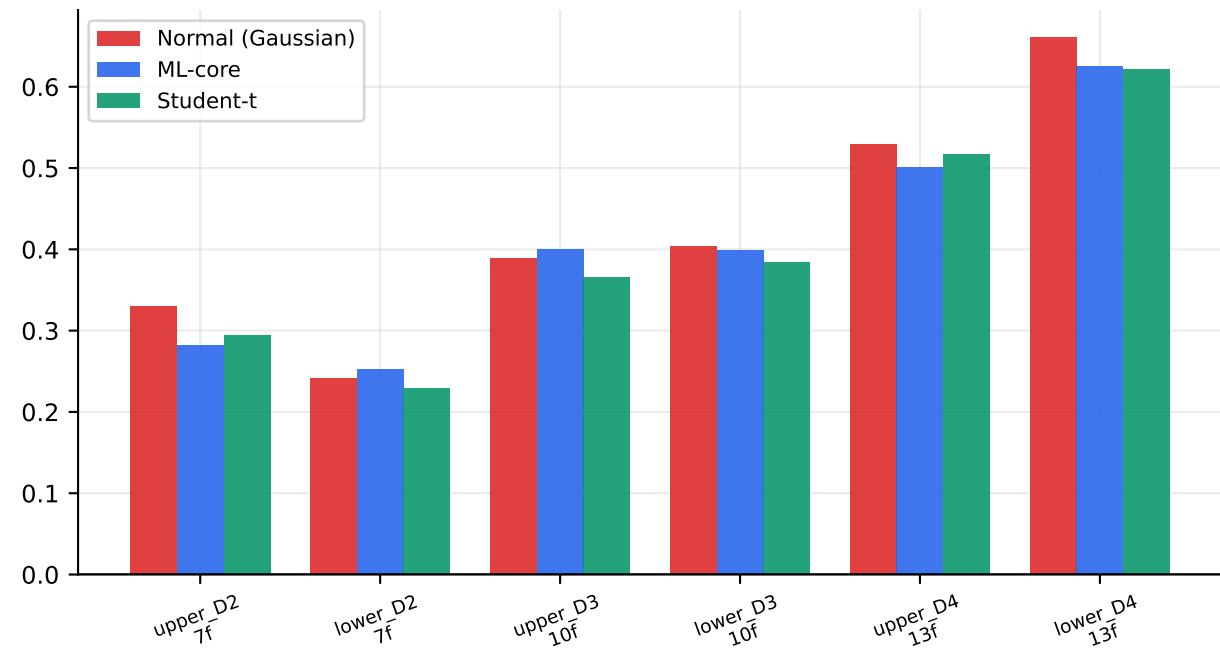


ML-core vs Normal vs Student-t — identical inputs

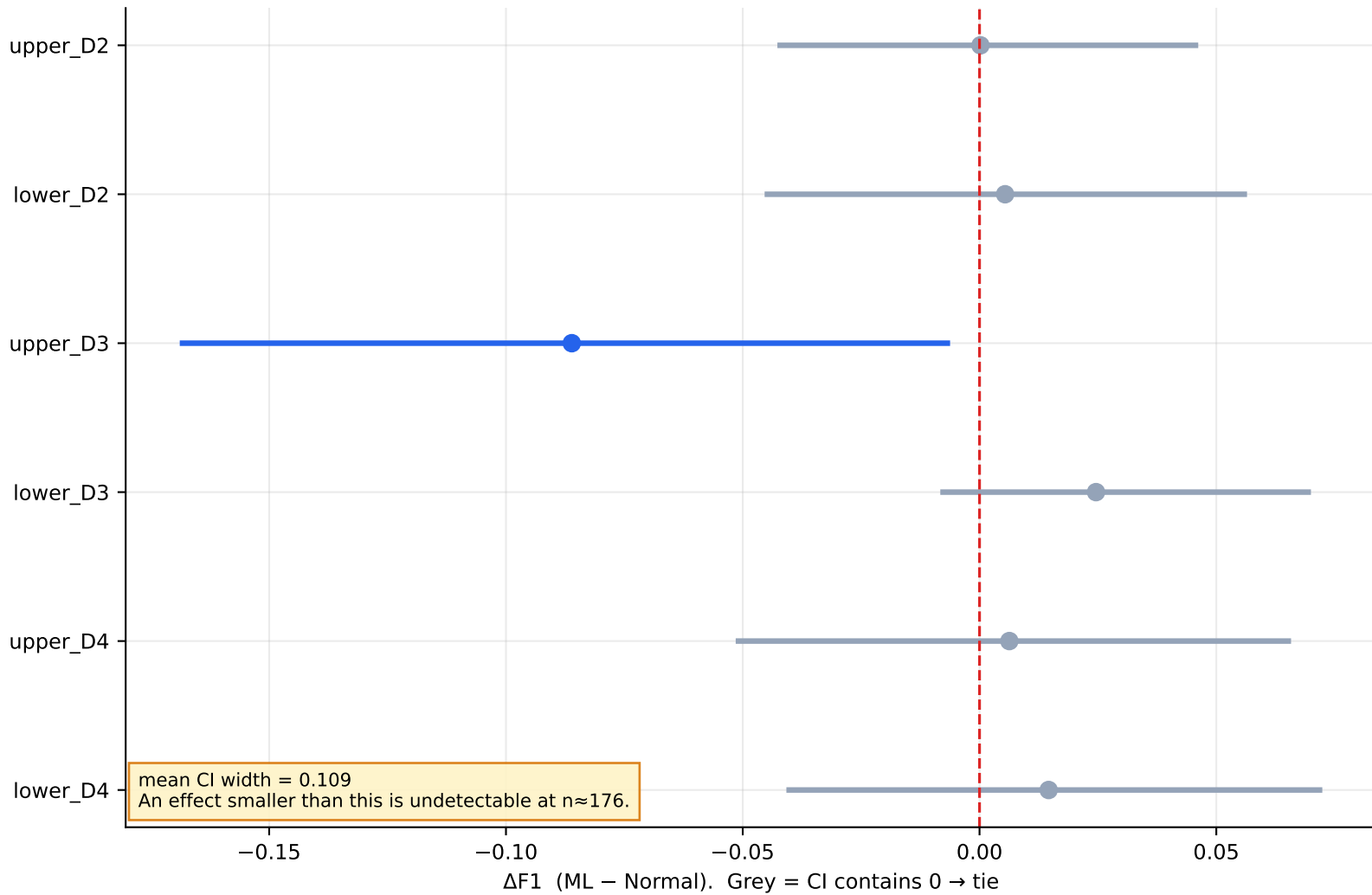
F1 (threshold-dependent)



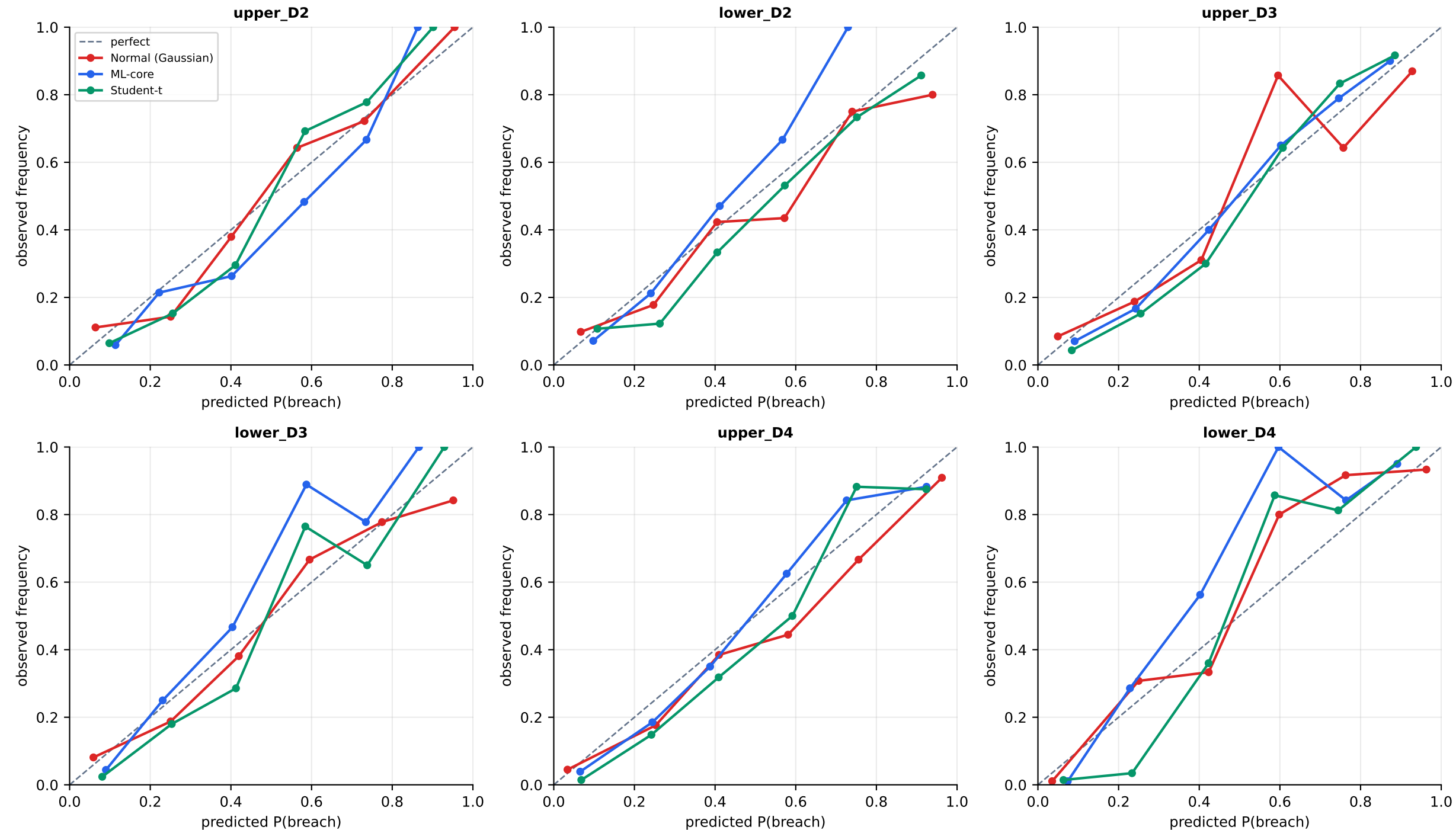
Brier skill (threshold-free)



Paired bootstrap, 95% CI — resampling cycles

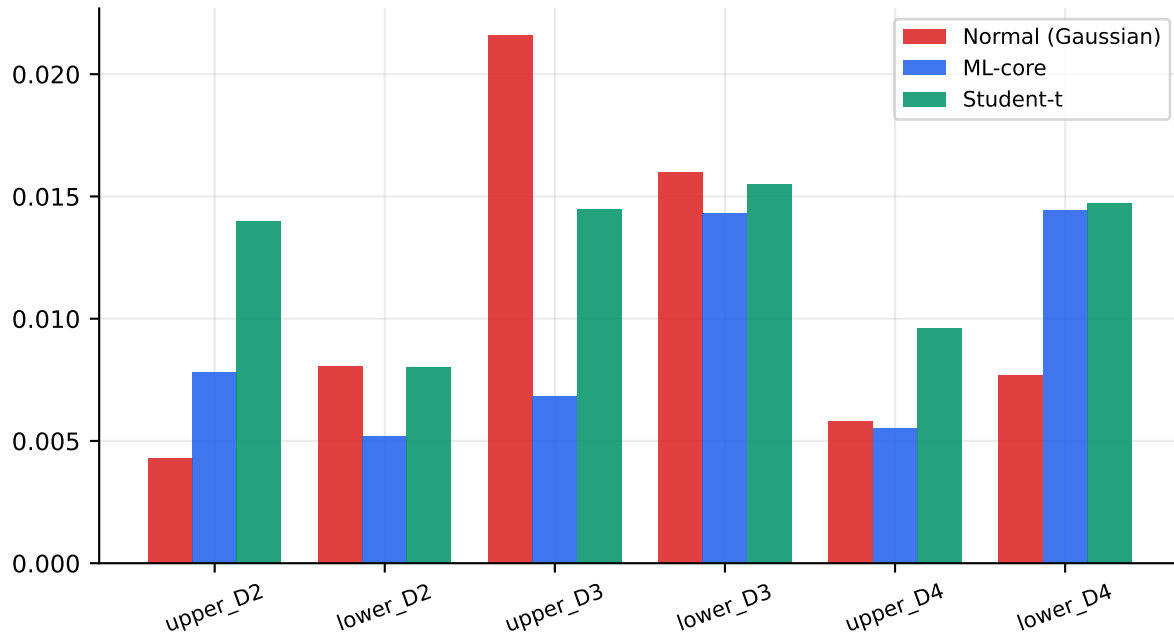


Reliability — above the diagonal = under-predicting breaches

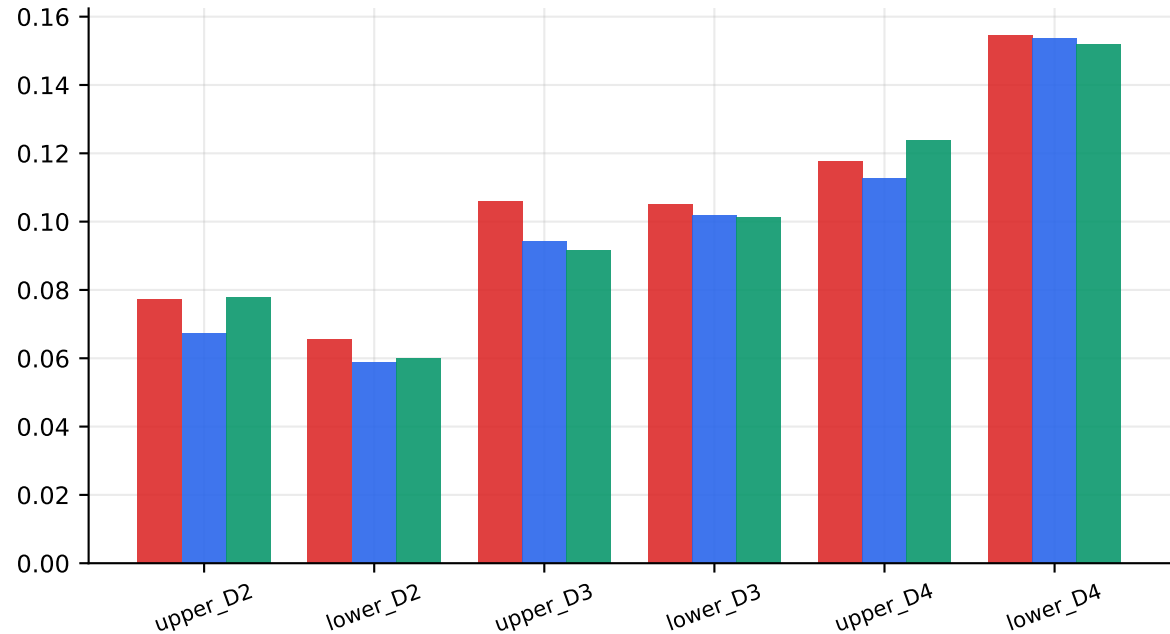


Brier decomposition: $BS = \text{reliability} - \text{resolution} + \text{uncertainty}$
Large reliability for Normal $\Rightarrow$ exploitable structure. Low resolution for all $\Rightarrow$ Bayes floor.

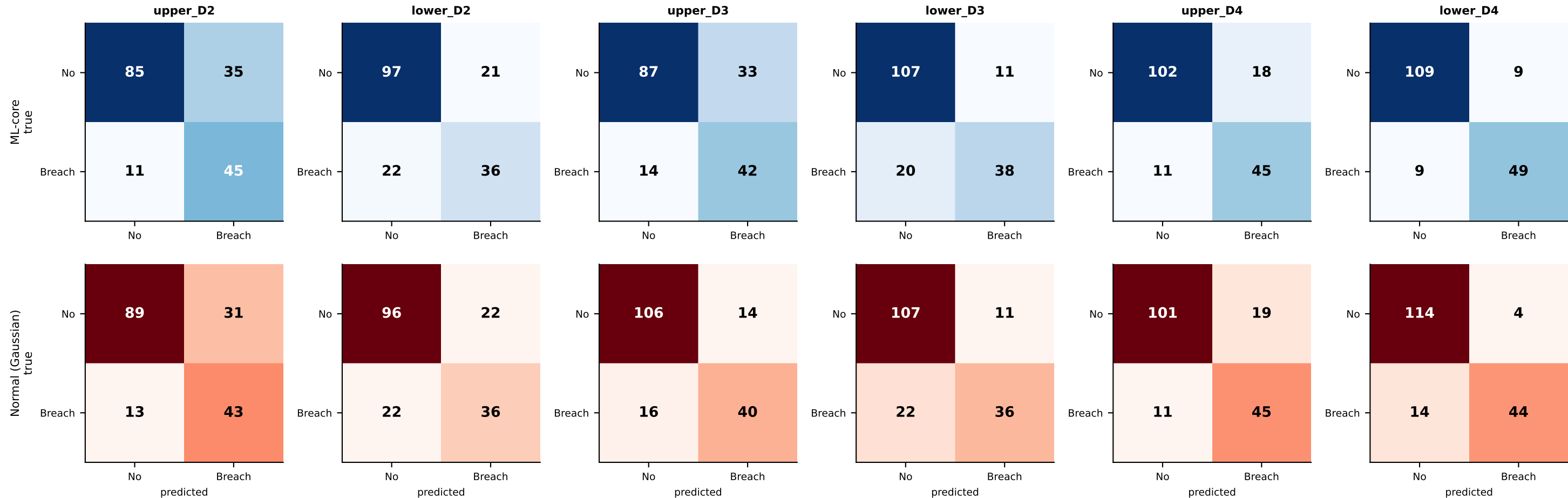
Reliability (miscalibration — LOWER is better)



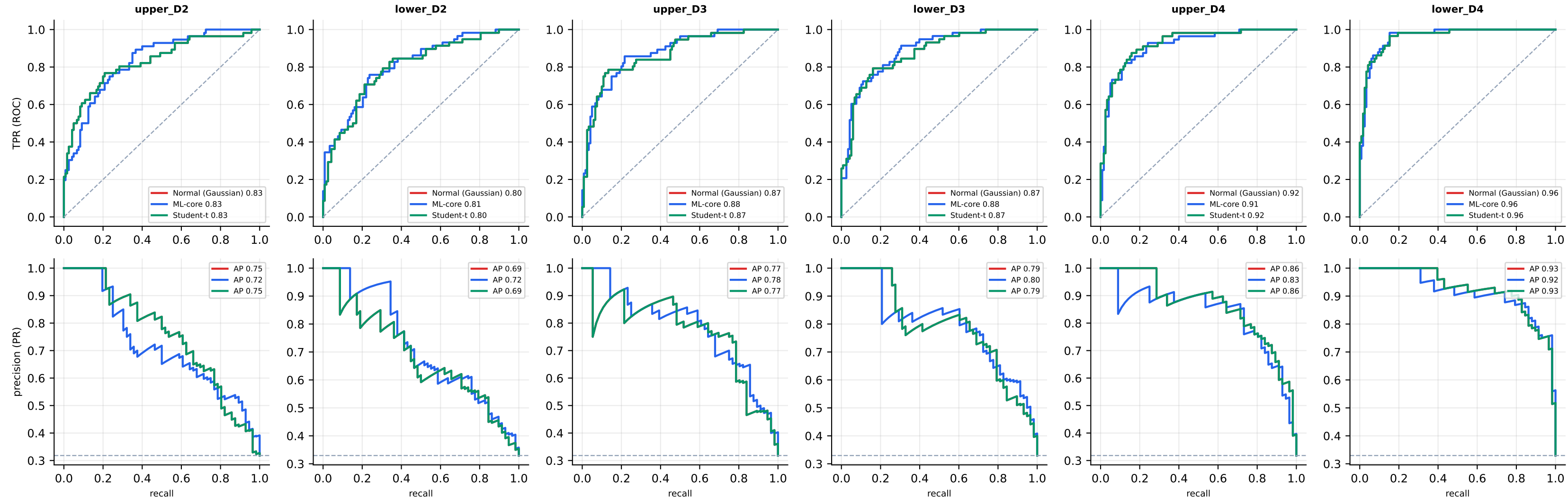
Resolution (discrimination — HIGHER is better)



Confusion matrices

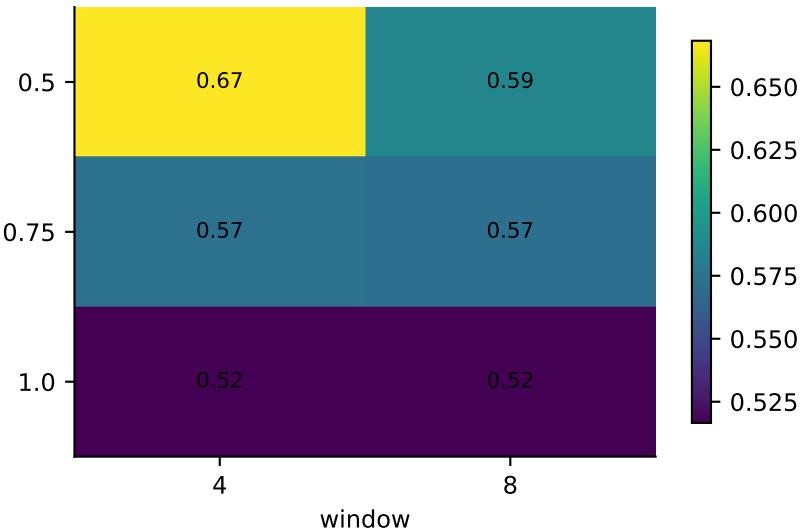


ROC and Precision-Recall. PR is the honest curve for a rare positive class — the dashed line is the base rate.

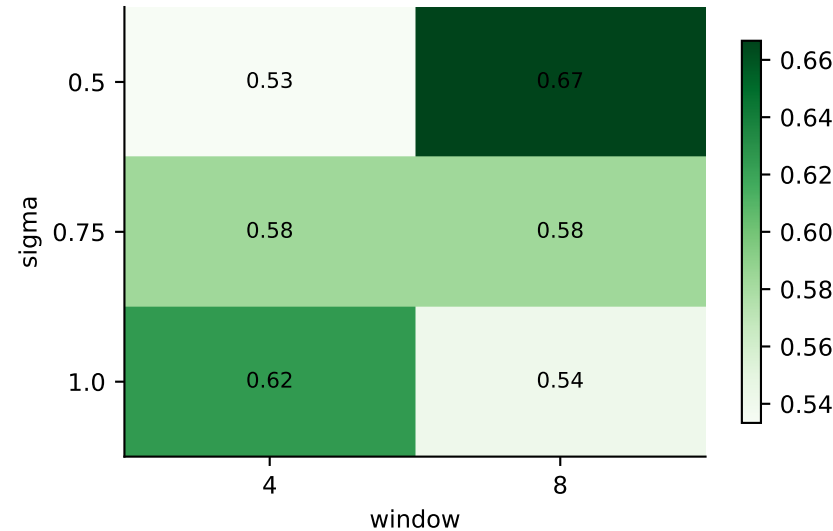


Config robustness. ⚠ The base rate changes with sigma (the band IS the label),
so $\Delta F1$ is comparable **WITHIN** a cell but not **ACROSS** sigma levels.

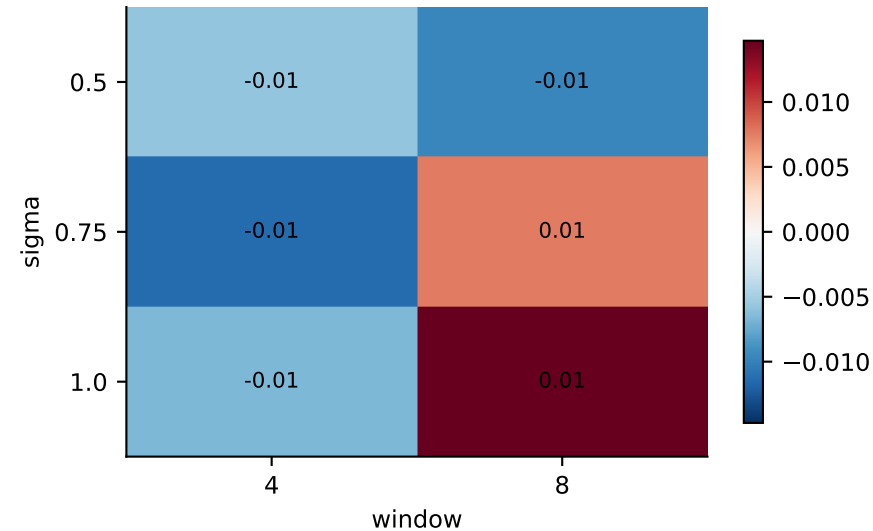
best oof_score (consistent only)



consistency pass rate



mean $\Delta F1$ (ML – Normal)



Error overlap — a large 'both wrong' block means the two contestants fail on the SAME cycles, so no hybrid of them can help.

